

Basic Principles of Medicine 2

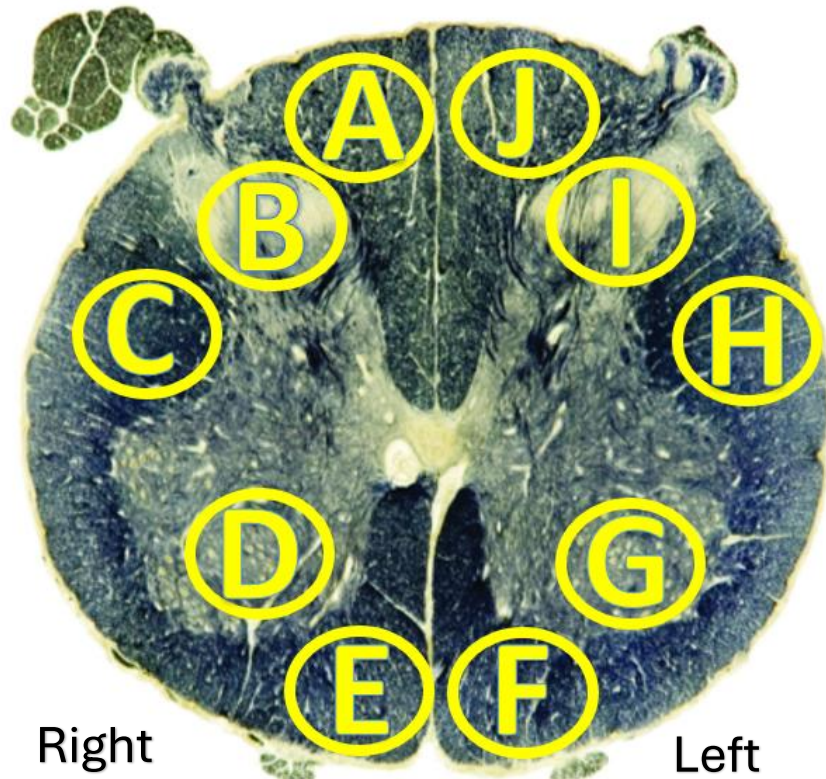
Nervous System and Behavioral Sciences

IMCQ 6

St. George's University

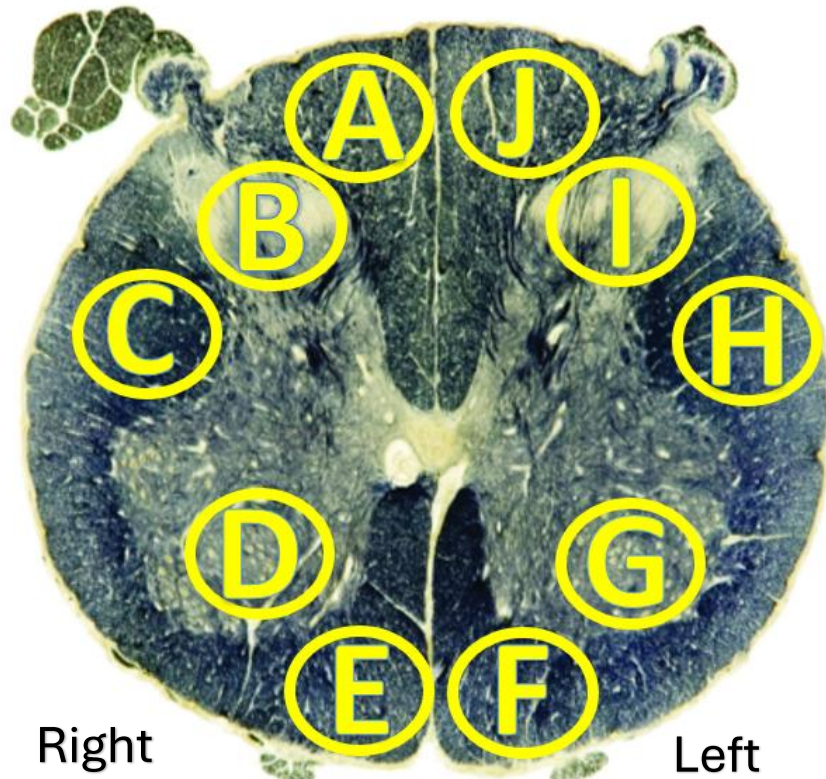
1. A 5-year-old boy is brought to the physician by his mother because of a 3-day history of difficulty walking. Two weeks ago, he had a viral infection that resolved spontaneously. He recently immigrated to the USA. Vaccinations are not up-to-date. Neurologic examination shows decreased muscle strength and tone of the right leg. Deep tendon reflexes are absent in the right lower extremity. Sensation is intact. Which of the following labeled sites in the photograph of a cross-section of the spinal cord is most likely damaged in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G
- H. H
- I. I
- J. J



1. A 5-year-old boy is brought to the physician by his mother because of a 3-day history of difficulty walking. Two weeks ago, he had a viral infection that resolved spontaneously. He recently immigrated to the USA. Vaccinations are not up-to-date. Neurologic examination shows decreased muscle strength and tone of the right leg. Deep tendon reflexes are absent in the right lower extremity. Sensation is intact. Which of the following labeled sites in the photograph of a cross-section of the spinal cord is most likely damaged in this patient?

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SOM.MK.II.BPM2.4.NB.2.NSCI.0173 Characterize lesions of the cell bodies or axons forming part of the motor unit
SOM.MK.I.BPM2.4.NB.2.NSCI.0260 Integrate knowledge of common manifestations of lower motor neuron lesions.
SOM.MK.I.BPM2.4.NB.2.NSCI.0261 Integrate knowledge of upper and lower motor neuronal signs with lesions affecting central and peripheral nervous sites

2. A 29-year-old woman comes to the physician because of a 2-day history of progressive weakness of the legs and tingling in both feet. Prior to her current symptoms, three weeks ago, she had a cough, fever, and malaise that resolved without treatment. Muscle strength is 2/5 in the lower extremities. Knee and ankle deep tendon reflexes are hypoactive. A lumbar puncture is performed. Cerebrospinal fluid (CSF) analysis is most likely to show which of the following sets of findings in this patient?

	Protein	White blood cell count
A	Decreased	Decreased
B	Increased	Increased
C	Increased	Within reference range
D	Within reference range	Increased
E	Within reference range	Within reference range

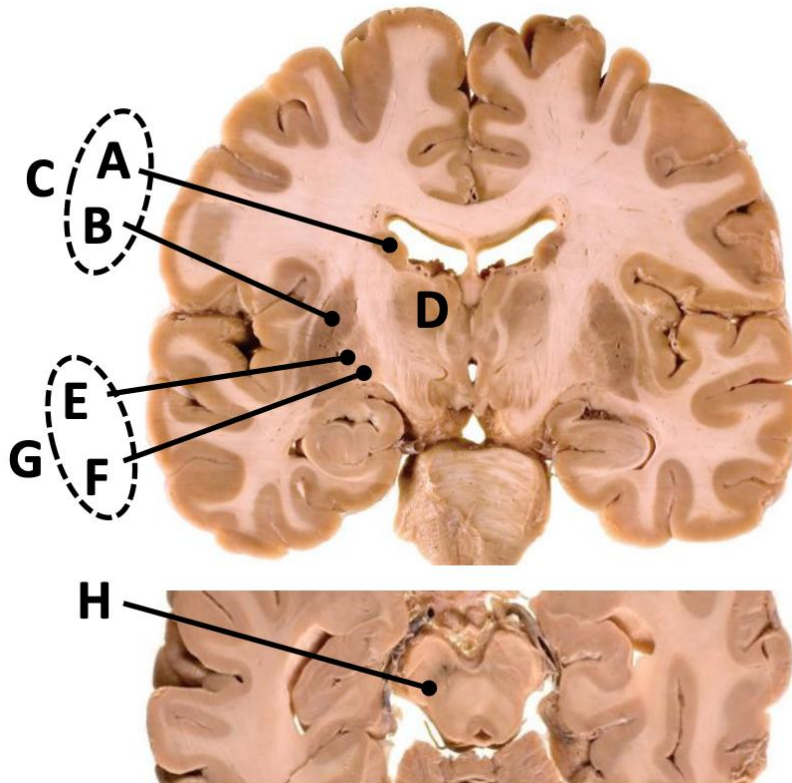
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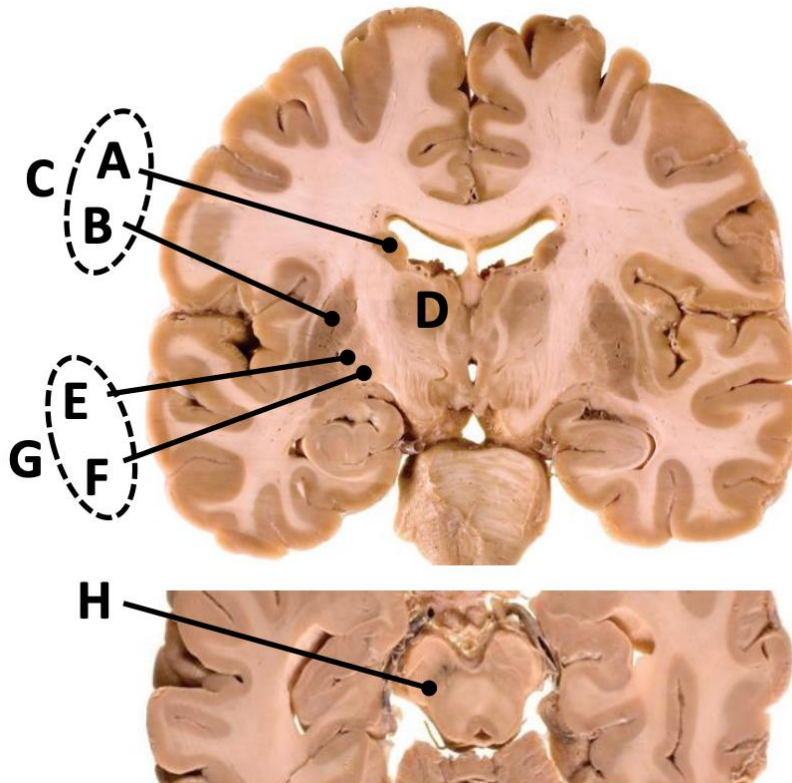
3. A 59-year-old man comes to the physician because of a 2-week history of rhythmic shaking of his left hand. Physical examination shows a pill-rolling tremor on the left side at rest. His facial expression is limited, his arms do not swing when he walks, and his gait is shuffling. His movement is slow and diminished. Neurologic examination shows cogwheel rigidity, which is worse on his left upper extremity. The neurologic findings in this patient are most likely due to diminished dopamine release into which of the following labeled structures in the photographs of a normal brain?

- A. A
- B. B
- C. C
- D. D
- E. E
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- E. E
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SOM.MK.I.BPM2.1.NB.1.NSCI.0187 Identify major components of the basal ganglia circuitry and general principles governing their functional interactions
SOM.MK.I.BPM2.1.NB.1.NSCI.0190 Relate the nature and causes of Parkinsonism to the direct and indirect motor loops

4. A 71-year-old woman comes to the physician because of a 3-day history of headaches and difficulty walking. Her vital signs are within normal limits. Neurologic examination shows gait ataxia with lurching to the left side and ataxia of the left upper and lower limbs. MRI of the brain shows a lesion in the cerebellum. Which of the following regions of the cerebellum is most likely affected in this patient?

- A. Left lateral hemisphere
- B. Left paravermis
- C. Left vermis
- D. Right lateral hemisphere
- E. Right paravermis
- F. Right vermis

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- C. Left vermis
- D. Right lateral hemisphere
- E. Right paravermis
- F. Right vermis

5. A 51-year-old man comes to the physician because of a 3-week history of problems with his vision. He has a history of hypertension. His blood pressure is 150/90 mm Hg; other vital signs are within normal limits. MRI of the brain shows a lesion in the left temporal lobe. Which of the following is the most likely visual field defect in this patient?

- A. Right monocular blindness
- B. Left monocular blindness
- C. Bitemporal hemianopia
- D. Superior left quadrantanopia
- E. Superior right quadrantanopia
- F. Inferior left quadrantanopia
- G. Inferior right quadrantanopia

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- G. Inferior right quadrantanopia

6. A 53-year-old man comes to the physician because of a 5-month history of vision changes and headaches. The patient has been squinting and has blurry vision when reading textbooks up close. He often holds reading materials at arm's length to see more clearly. He has no history of major medical illness and takes no medications. Physical examination shows equal and reactive pupils. Extraocular movements are intact. Visual field testing shows no deficits. Near visual acuity is 20/70 bilaterally. Fundusoscopic examination shows no abnormalities. Which of the following lenses shown is most appropriate to manage this patient?

A.
B.
C.
D.
E.



A



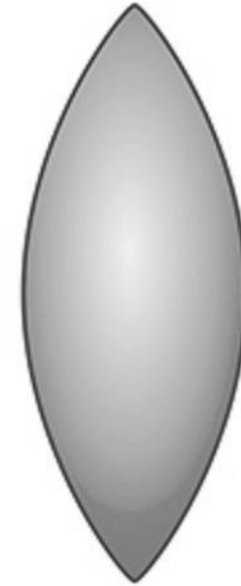
B



C

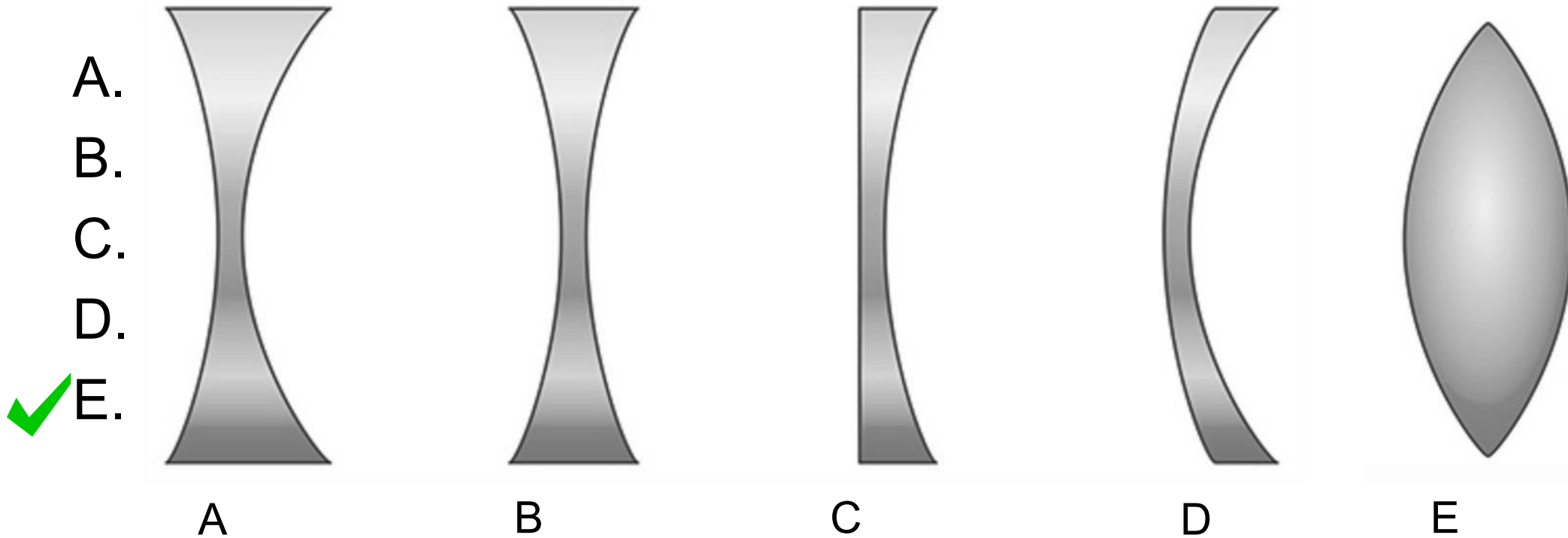


D



E

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7. A 43-year-old man comes to the physician because of a 1-week history of headaches and double vision. H-test shows normal gaze to the right, severely restricted adduction of the right eye on gaze to the left, normal convergence, and normal primary gaze. Which of the following lesions is most likely present in this patient?

- A. Left MLF
- B. Right MLF
- C. Left oculomotor
- D. Right oculomotor
- E. Left PPRF
- F. Right PPRF

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- ✓ B. Right MLF
- C. Left oculomotor
- D. Right oculomotor
- E. Left PPRF
- F. Right PPRF

8. A 69-year-old man comes to the physician because of a 2-day history of headache and dizziness. Neurologic examination shows truncal ataxia and impaired tandem walking. MRI of the brain shows a lesion in the flocculonodular lobe of the cerebellum. Which of the following additional findings is most likely in this patient?

- A. Dysarthria
- B. Dysdiadochokinesia
- C. Dysmetria
- D. Intention tremor
- E. Nystagmus

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- B. Dysdiadochokinesia
- C. Dysmetria
- D. Intention tremor
- ✓ E. Nystagmus

9. A 27-year-old man comes to the emergency department because of a 2-hour history of clumsiness on the left side of his body. He is agitated and says that he has smoked methamphetamine. Neurologic examination shows normal strength and somatosensory function. Cranial nerves are intact, although his pupils are dilated. Past-pointing and intention tremors are noted on the left. Which of the following structures is most likely damaged?

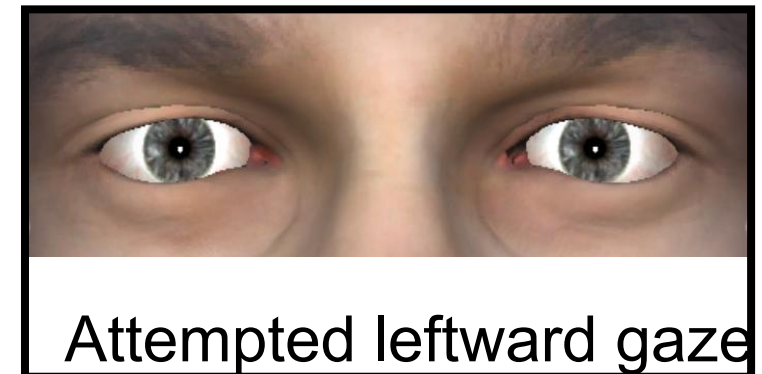
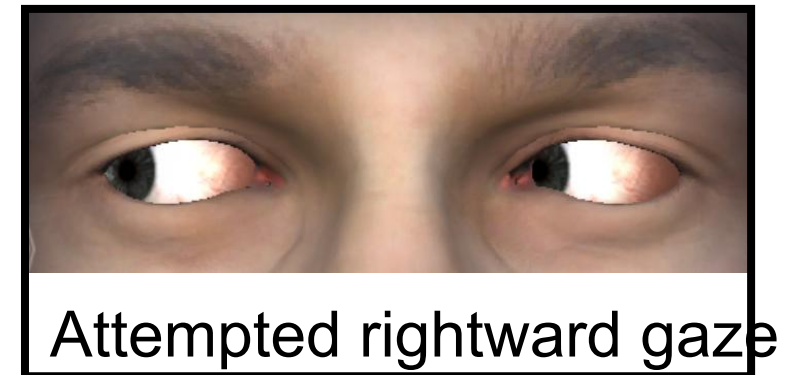
- A. Right cerebellum
- B. Left cerebellum
- C. Right striatum
- D. Left striatum
- E. Right substantia nigra
- F. Left substantia nigra
- G. Right subthalamus
- H. Left subthalamus

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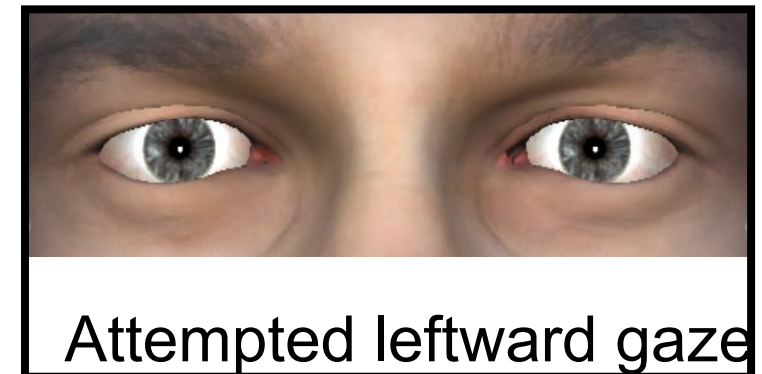
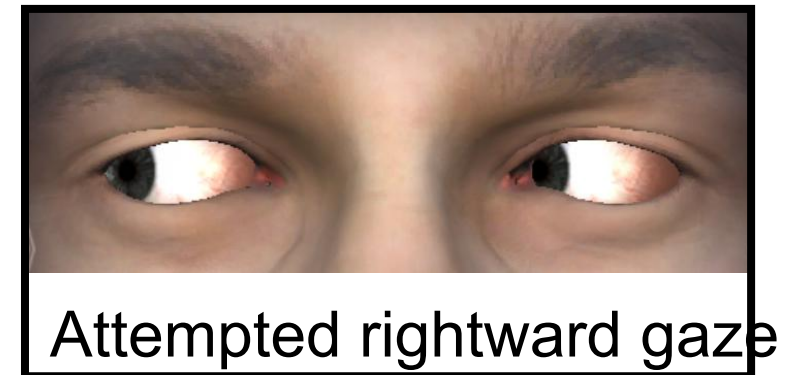
10. A 56-year-old man comes to the physician because of a 3-month history of visual problems. Direct and consensual pupillary light responses are bilaterally normal. Visual field testing shows intact vision in all four quadrants of the visual fields of each eye. Findings for primary gaze and attempted horizontal gaze are shown. Branch occlusion of which of the following arteries is most likely in this patient?

- A. Anterior cerebral
- B. Basilar
- C. Posterior cerebral
- D. Posterior inferior cerebellar
- E. Superior cerebellar
- F. Vertebral



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- E. Superior cerebellar
- F. Vertebral



11. A 27-year-old man comes to the physician because of a 7-month history of progressive vision changes. He has had difficulty seeing at night and navigating around the office. He does not have eye pain or floaters. He has no history of major medical illness and takes no medications. His mother and maternal grandfather have vision problems. He eats a well-balanced diet. He does not smoke cigarettes, drink alcoholic beverages, or use illicit drugs. Physical examination shows pupils are equal and reactive to light. Confrontational visual field testing shows concentric peripheral field loss. The results of a funduscopic examination are shown. Laboratory studies show decreased total serum cholesterol. Which of the following is the most likely underlying cause of this patient's condition?

- A. Age-related deterioration of the macula
- B. Chronic hyperglycemia-induced microvascular injury
- C. Hereditary dystrophy of the retina
- D. High intraocular pressure leading to ganglion cell death
- E. Vitamin A deficiency affecting rhodopsin synthesis



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12. A 60-year-old man comes to the physician because of a 3-week history of intermittent visual disturbances. He also has had headaches, nausea, and vomiting during this time. He has smoked one pack of cigarettes daily for 35 years. Visual field testing shows an enlargement of the physiological blind spot bilaterally. The results of a funduscopic examination are shown. The most likely cause of this patient's findings is an alteration in which of the following?

- A. Aqueous outflow
- B. Choroid attachment to the retina
- C. Intracranial pressure
- D. Optic nerve myelination
- E. Retinal pigment epithelium integrity



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13. A 65-year-old woman comes to the physician because of a 3-month history of hearing difficulty. Neurologic examination shows AC>BC for both right and left ears, but times for AC and BC are diminished for the left as compared to the right. Which of the following structures is most likely lesioned?

- A. Left inferior colliculus
- B. Left parietal lobe
- C. Left temporal lobe
- D. Left tympanic membrane
- E. Left vestibulocochlear nerve
- F. Right inferior colliculus
- G. Right parietal lobe
- H. Right temporal lobe
- I. Right tympanic membrane
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14. A 62-year-old man comes to the physician because of a 7-month history of problems hearing. The patient can repeat two-syllable words whispered on the left, but not on the right. Physical examination shows cerumen impaction in the right ear. Which of the following set of additional findings are most likely during neurologic examination?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G
- H. H
- I. I

Choice	Weber's test	Rinne's - Left	Rinne's - Right
A	Symmetrical	AC>BC	AC>BC
B	Left lateralization	AC>BC	AC>BC
C	Left lateralization	AC>BC	BC>AC
D	Left lateralization	BC>AC	AC>BC
E	Left lateralization	BC>AC	BC>AC
F	Right lateralization	AC>BC	AC>BC
G	Right lateralization	AC>BC	BC>AC
H	Right lateralization	BC>AC	AC>BC
I	Right lateralization	BC>AC	BC>AC

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C	Left lateralization	AC>BC	BC>AC
D	Left lateralization	BC>AC	AC>BC
E	Left lateralization	BC>AC	BC>AC
F	Right lateralization	AC>BC	AC>BC
G	Right lateralization	AC>BC	BC>AC
H	Right lateralization	BC>AC	AC>BC
I	Right lateralization	BC>AC	BC>AC